ENTITIES AND RELATIONS

1. Threat actor
2. Threat group
3. Malware
4. Attack technique
5. Vulnerability
6. Software
7. Tool
8. Organization
9. Attack pattern
10. File/hash
11. Network infrastructure
12. Asset/system
13. Security team
14. Time
15. Location

Mapped Relations (Source ID → Relation → Target ID)

Threat Actor & Threat Group Relations

- 1 → alias-of → 2 (or 2 → alias-of → 1)
- 1 → uses → 3 (Threat actor uses malware)
- 1 → uses → 4 (Threat actor uses attack technique)
- 1 → uses → 7 (Threat actor uses tool)
- 1 → uses → 9 (Threat actor uses attack pattern)
- 1 → uses → 6 (Threat actor uses software)
- 2 → uses → 3 (Threat group uses malware)
- 2 → uses → 4 (Threat group uses attack technique)

- 2 → uses → 7 (Threat group uses tool)
- 2 → uses → 9 (Threat group uses attack pattern)
- 1 → targets → 8 (Threat actor targets organization)
- 1 → targets → 12 (Threat actor targets asset/system)
- 2 → targets → 8 (Threat group targets organization)
- 2 → targets → 12 (Threat group targets asset/system)
- 1 → operates-in → 15 (Threat actor operates in location)
- 2 → operates-in → 15 (Threat group operates in location)
- 1 → active-at → 14 (Threat actor active at time)
- 2 → active-at → 14 (Threat group active at time)
- 1 → originated-from → 15
- 2 → originated-from → 15

Malware, Tool, & Software Relations

- 3 → variant-of → 3
- 7 → variant-of → 7
- 3 → drops → 10 (Malware drops file/hash)
- 3 → exploits → 5 (Malware exploits vulnerability)
- 3 → affects → 6 (Malware affects software)
- 3 → affects → 12 (Malware affects asset/system)
- 7 → exploits → 5
- 3 → communicates-with → 11 (Malware communicates with network infrastructure)
- 7 → communicates-with → 11
- 6 → contains → 5 (Software contains vulnerability)
- 6 → runs-on → 12 (Software runs on asset/system)

Attack Technique & Attack Pattern Relations

- 4 → sub_technique_of → 4
- 9 → variant-of → 9
- 4 → uses → 7 (Attack technique uses tool)

- 4 → uses → 3 (Attack technique uses malware)
- 9 → uses → 7
- 4 → targets → 5 (Attack technique targets vulnerability)
- 4 → targets → 12 (Attack technique targets asset/system)
- 4 → mitigated-by → 13 (Attack technique mitigated by security team)

Vulnerability & Asset Relations

- 5 → affects → 6 (Vulnerability affects software)
- 5 → affects → 12 (Vulnerability affects asset/system)
- 5 → remediated-by → 6 (Vulnerability remediated by software patch)
- 12 → belongs-to → 8 (Asset/system belongs to organization)
- 12 → located-in → 15 (Asset/system located in location)
- 12 → monitored-by → 13 (Asset/system monitored by security team)

File/Hash & Network Infrastructure Relations

- 10 → indicates → 3 (File/hash indicates malware)
- 10 → indicates → 1 (File/hash indicates threat actor)
- 10 → downloads-from → 11 (File downloads from network infrastructure)
- 11 → belongs-to → 8 (Network infrastructure belongs to organization)
- 11 → located-in → 15 (Network infrastructure located in location)
- 11 → utilized-by → 1 (Network infrastructure utilized by threat actor)
- 11 → utilized-by → 2 (Network infrastructure utilized by threat group)
- 11 → utilized-by → 3 (Network infrastructure utilized by malware)

Security Team, Time, & Location Context Relations

- 13 → investigates → 1 (Security team investigates threat actor)
- 13 → investigates → 2 (Security team investigates threat group)
- 13 → detects → 3 (Security team detects malware)
- 13 → detects → 5 (Security team detects vulnerability)
- 13 → detects → 9 (Security team detects attack pattern)
- 13 → mitigates → 5 (Security team mitigates vulnerability)

- 13 → patches → 12 (Security team patches asset/system)
- Any Entity → occurs-at → 14 (Linked to time)
- Any Entity → located-in → 15 (Linked to location)